



University of St.Gallen

Institute of Computer Science

Process KPIs

Process Mining

Session 3



From insight to impact.

Outline

Session 1: Introduction

- *Event Logs, Event Log Inspection, First Exploration*

Session 2: Exploratory Log Analysis

- *Schema Analysis, Visualizations, Log Enrichments*

Session 3: Process KPIs

- *Process Outcomes, Dashboards*

Session 4: Process Discovery and –Modeling

- *Variants, Process Maps, BPMN, Petri Nets*

Assignment 1 Part 1 due

Session 5: Conformance Checking

- *Business Rules and Normative Process Models*

Session 6: Performance Mining

- *Load and Demand, Bottlenecks*

Session 7: Deviance Diagnostics

- *Explanations and Rule Mining*

Assignment 1 Part 2 due

Process KPIs

1. Concepts and Overview
2. Process Outcome Analysis
3. Dashboards

Process KPIs

KPI = Key Performance Indicator = A measure for effectiveness or efficiency of a business process.

Process Performance

Cost (vs. Benefit)

ROI

Return on Investment

Cost per Case

Resource Utilization

Time

Cycle Time

Total time taken to complete a process from start to finish.

Throughput

The number of cases processed within a given time period.

Quality

Process Quality

Error Rate

Rework Rate

Percentage of work that needs to be redone due to errors.

Customer Satisfaction

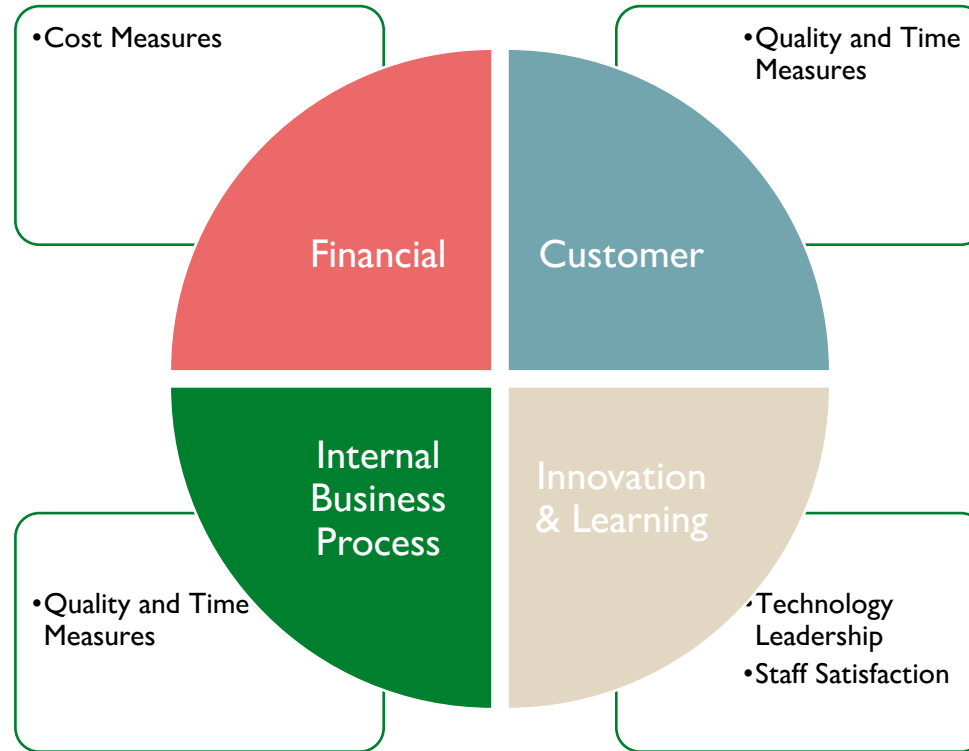
Customer Complaints

NPS

Net Promoter Score (Willingness of customer to recommend service.)

and more ...

Balanced Scorecard



More ...

Process Compliance

- *Compliance Rate*: The extent to which the process adheres to regulatory or internal standards. A high compliance rate is crucial in industries with strict regulations.
- *Audit Findings*: The number and severity of issues found during audits.

Process Flexibility

- *Changeover Time*: The time required to switch from producing one product or service to another. Shorter changeover times indicate greater flexibility.
- *Adaptability Index*: Measures how quickly the process can adapt to changes in demand or conditions. Higher adaptability is crucial in dynamic environments.

Process Sustainability

- *Waste Reduction*: The amount of waste produced by the process. Lower waste levels indicate a more sustainable process.
- *Energy Efficiency*: Measures the amount of energy consumed relative to the output produced. Higher energy efficiency reflects better sustainability practices.

Innovation and Improvement

- *Continuous Improvement Rate*: The frequency and impact of improvements made to the process. Higher rates suggest a strong culture of innovation.
- *Innovation Index*: Measures the process' ability to incorporate new technologies or methods to enhance performance.

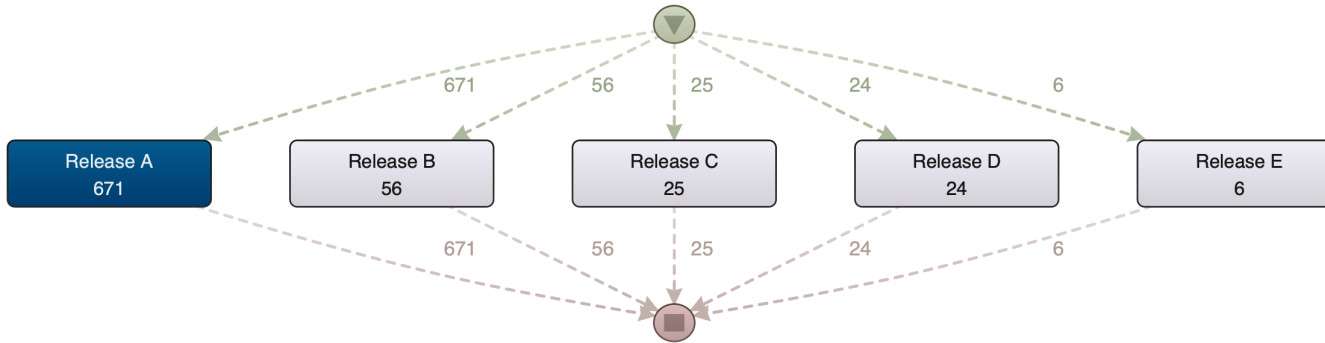
Process Transparency

- *Visibility Index*: The degree to which process performance can be monitored and tracked in real-time. Higher visibility allows for quicker decision-making and problem resolution.

Process Outcome Analysis

Foundational for Cost vs. Benefit Analysis

Use Disco for Outcome Distribution: Simple Case: Each Outcome is Represented as Activity



What is the problem here?

Attribute
Removes events by attribute

Filter by: Activity

Filtering mode:
☒ Keep selected
☐ Mandatory
☐ Forbidden

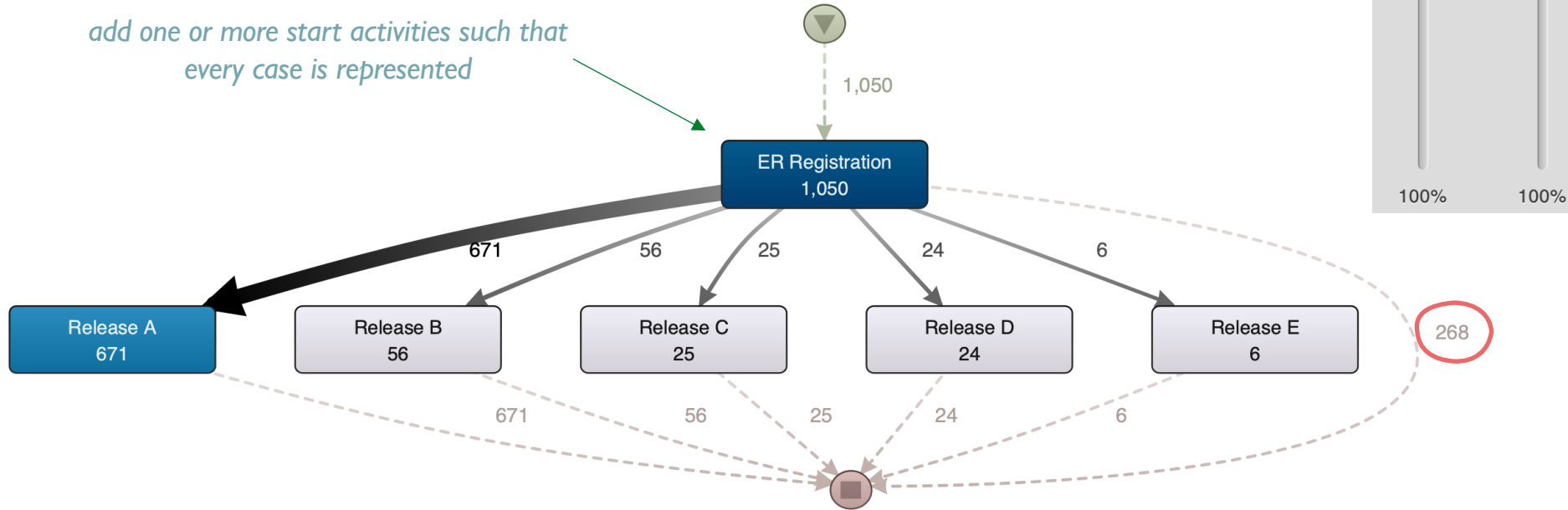
This filter removes all events that do not have one of the selected values.

Event values: (5 of 16 selected)

☒	Admission IC
☒	Admission NC
☒	CRP
☒	ER Registration
☒	ER Sepsis Triage
☒	ER Triage
☒	IV Antibiotics
☒	IV Liquid
☒	LacticAcid
☒	Leucocytes
☒	Release A
☒	Release B
☒	Release C
☒	Release D
☒	Release E
☒	Return ER

Case Outcome Distribution

add one or more start activities such that every case is represented



Process Outcomes (= qualitative process KPI, cf. sepsis_v2.ipynb)

```
case_log['Release A'] = case_log['seq_variant'].apply(lambda x: 'Release A' in x)
case_log['Release B'] = case_log['seq_variant'].apply(lambda x: 'Release B' in x)
case_log['Release C'] = case_log['seq_variant'].apply(lambda x: 'Release C' in x)
case_log['Release D'] = case_log['seq_variant'].apply(lambda x: 'Release D' in x)
case_log['Release E'] = case_log['seq_variant'].apply(lambda x: 'Release E' in x)
```

- We can think of different possible outcomes of a case being **mutually exclusive**, e.g., successful, failure1, failure2, ...
 - e.g., *loan signed, loan declined, incomplete application documents, application canceled, etc.*
- First we create for each outcome, an **outcome indicator** = a boolean flag whether the outcome has occurred
- ...

	Release A	Release B	Release C	Release D	Release E
case:concept:name					
A	True	False	False	False	False
AA	False	False	False	False	False
AAA	True	False	False	False	False
AB	False	False	False	False	False
ABA	True	False	False	False	False
...	...	...	...	...	...

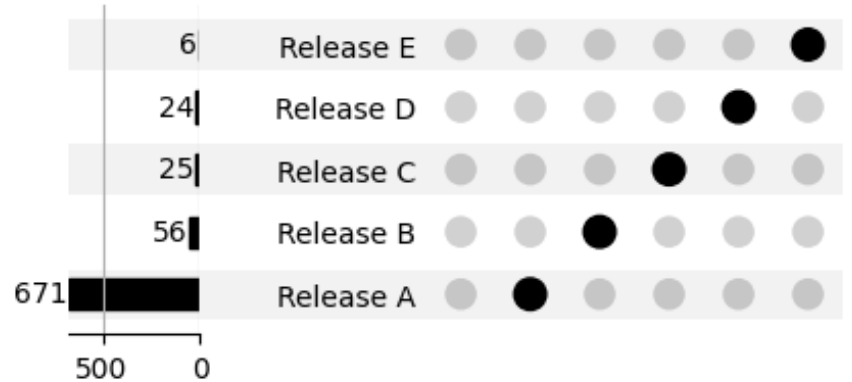
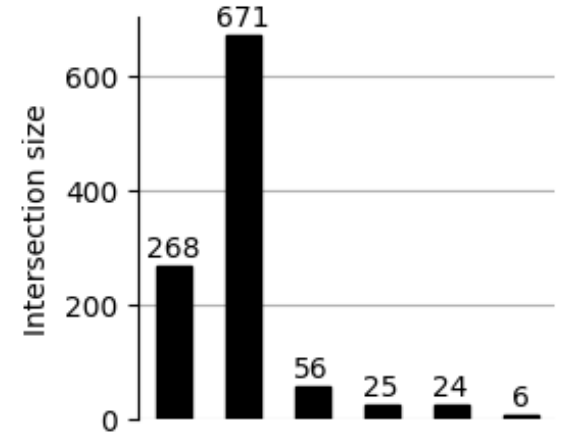
↑
single outcome indicator

UpSet Plot

```
from upsetplot import UpSet, from_indicators
```

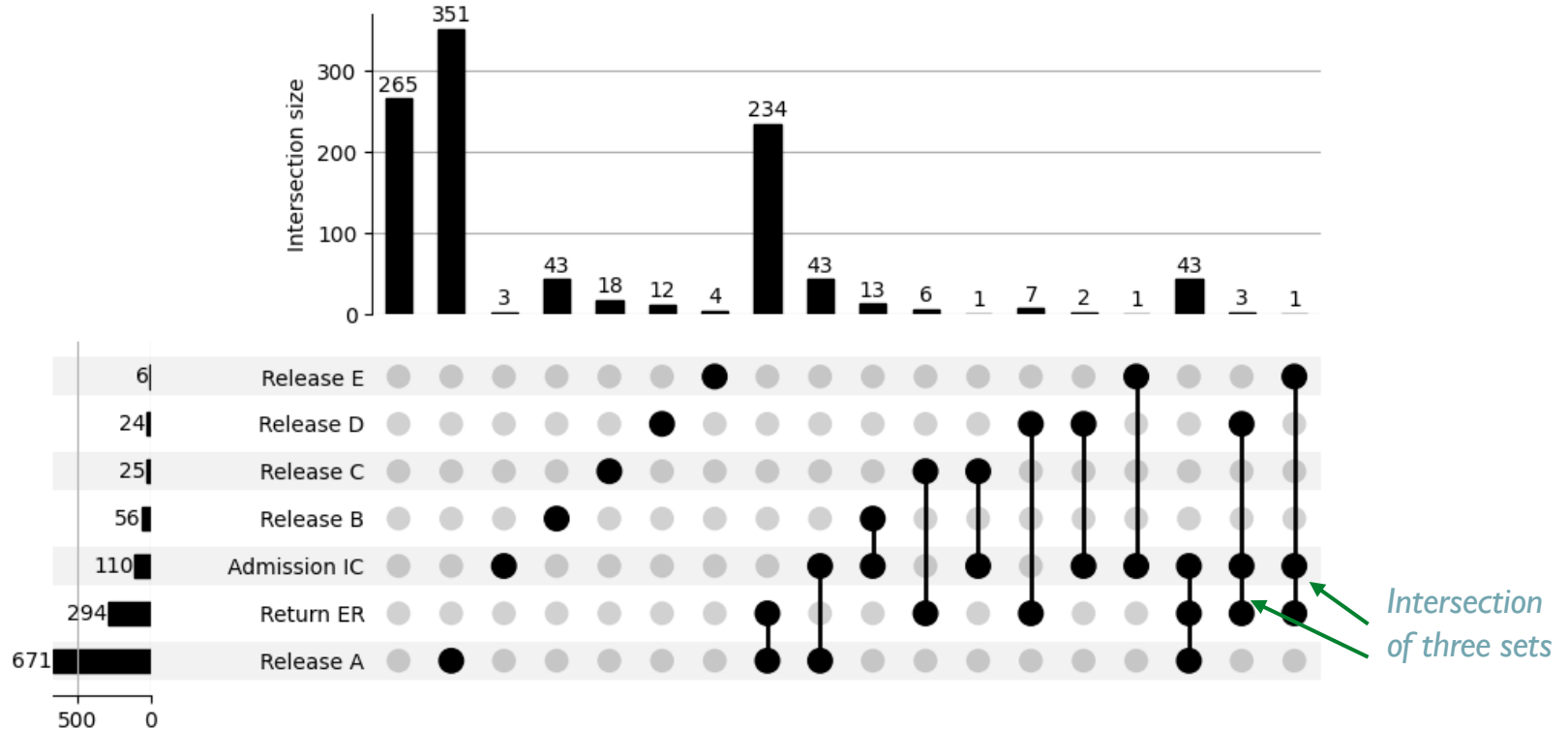
```
UpSet(from_indicators(['Release A', 'Release B', 'Release C',  
                      'Release D', 'Release E'],  
      data=case_log),  
      show_counts=True).plot()
```

- ...
- We then check whether the outcomes overlap
- Minor overlaps can be neglected after an investigation
- ...



More General UpSet Example

More info on UpSet [here](#).



Folding

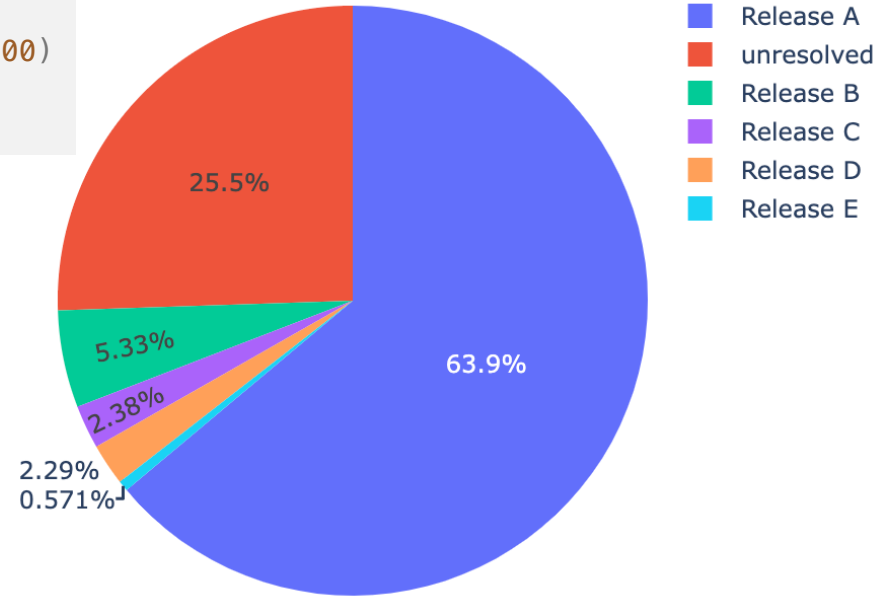
```
def fold_outcome(row, indicator_list=[]):  
    for outcome in indicator_list:  
        if row[outcome]:  
            return outcome  
    return 'unresolved'  
  
outcome_list = ['Release A', 'Release B', 'Release C', 'Release D', 'Release E']  
case_log['outcome'] = case_log.apply(fold_outcome, indicator_list=outcome_list, axis=1)
```

- ...
- when outcomes are proved to be **almost** mutually exclusive, we can fold the indicators into a categorical value *outcome*

	Release A	Release B	Release C	Release D	Release E	outcome
case:concept:name						
AO	True	False	False	False	False	Release A
AP	False	False	False	False	False	unresolved
AQ	True	False	False	False	False	Release A
AR	True	False	False	False	False	Release A
AS	False	True	False	False	False	Release B
AT	False	False	False	False	False	unresolved

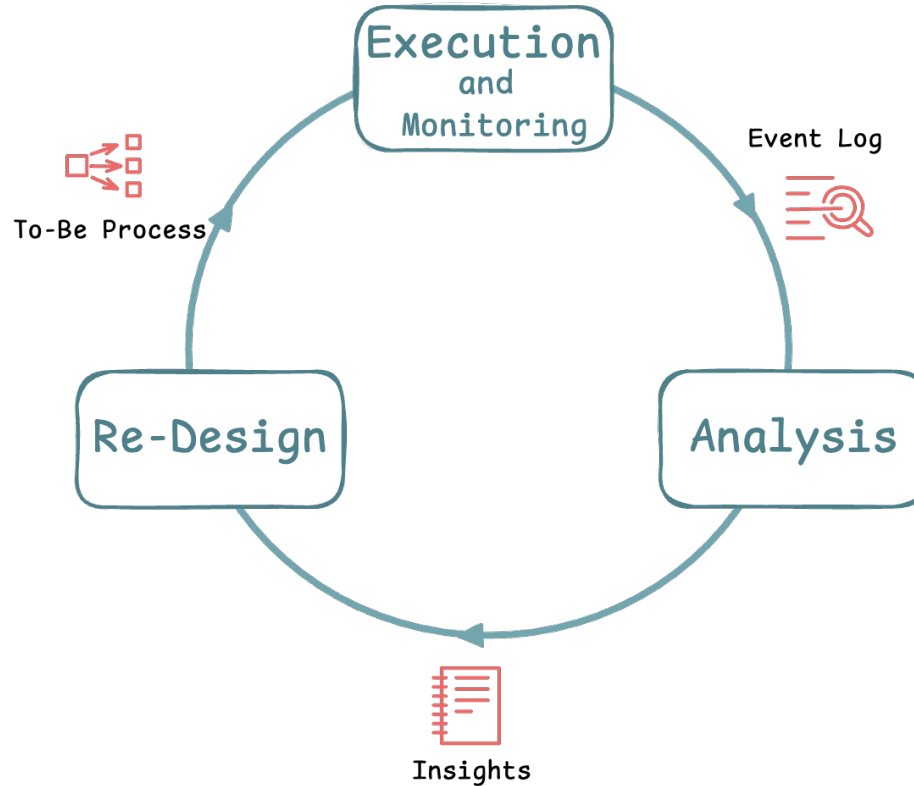
Plotting Outcomes

```
series = case_log['outcome'].value_counts()  
fig = px.pie(values=series, names=series.index, width=500)  
fig.show()
```



Process Dashboards

Monitoring in the Process Improvement Lifecycle



Process Overview

Start Date

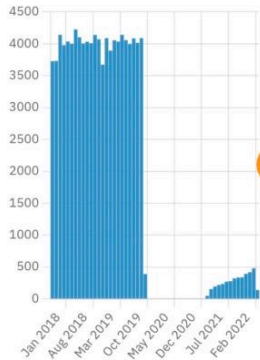
End Date

Off

Off

Reset all

1 Process Instance Usage



2 Overall Incident-Free Rate

95.76%

0% Target 99.5%

3 Automation Rate (<1 hour)

3.42%

0% Target 70%

4 Long-Running Process Instances

- Hiring Demo with unique correlation values
- Optimize DRI Process with loads of Variables
- Branch Analysis Process
- DG Pick Up Handling
- Customer Onboarding
- Multi-Instance Subprocess
- Book Request with no business key
- Multi Parallel Gateways



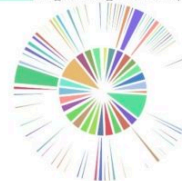
5 Automation Candidates

- Shipment File Preparation
- Accept Insurance Rejection
- Follow-up Insurance Booking
- Arrange Shipment Transfer to Destination
- Arrange Shipment Transfer to Destination
- Approve Result
- Verify approval
- Angaben prüfen
- Approve Invoice
- Checkout Book



6 Active Bottlenecks

- Incident that sometimes gets resolved
- No Incident
- Incident that stays open
- Check possibility to create an incident
- Adjust Inventory Data
- Create open template
- Export Shipment Completed
- AE0010 Start
- Services Handling
- Flight Arrangement Completed



Processes

Goals display data from instances ended in the last 30 days



Name

Owner

7 KPI: Time

8 KPI: Quality



Process
Document Check Handling

2 1



Process
Order Confirmation

1 2



Process
Embedded Subprocess

1 2



Process
Hiring Demo 5 Tenants

Add Owner

3

3

Report: Incident-Free Rate

Value: 100% Target: 99%

Report: Candidate Applications (30-day rolling)

Value: 425 Target: 50

Report: Hiring Rate

Value: 26.68% Target: 20%

On-Time Delivery rate ⓘ

68.3%

-0.80 pp

Avg. delay

6.5 days

Sales order items ⓘ

118K

Net order value ⓘ

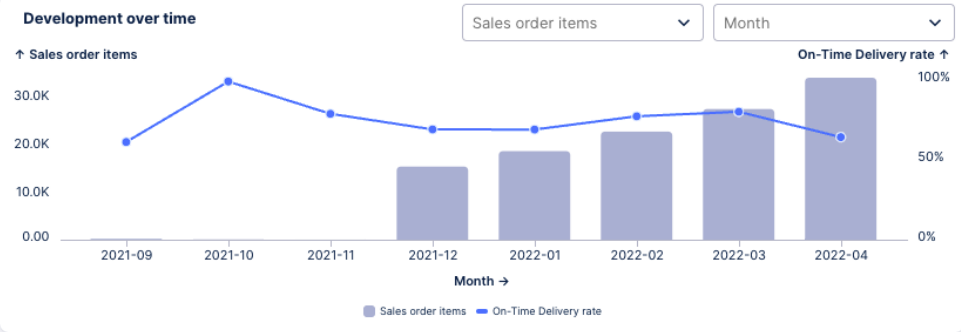
20.5B EUR

Process

Settings

Value Opportunities

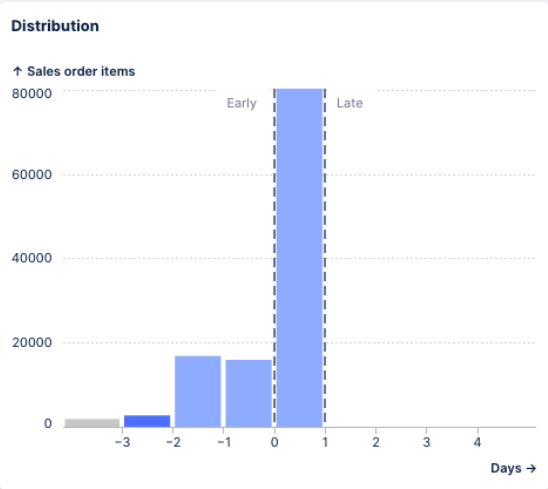
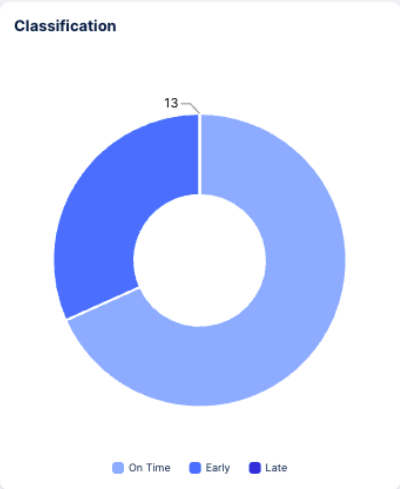
Value opportunity	Sales order items	Net order value ⓘ
Short requested lead time	118K	20.5B EUR
Insufficient material availability	5	5.01M EUR
Delivery block	7	19.4K EUR
Slow processing	3	13.6K EUR
Order change	3	136 EUR



Key Metrics by Dimension

Material

Material	Sales order items	Net order value ⓘ	On-Time Delivery rate
CEAT tyre	9K	18B EUR	68.3%
Tubeless tyre model 40...	5	1.8B EUR	100%
SP Sport 5000M DSST ...	5.27K	269M EUR	64.4%
Laser systems	6.82K	122M EUR	65%
Tube tyre model 115	5.27K	107M EUR	64.4%
Tubeless tyre model 40...	4.55K	35.8M EUR	70.6%
Tubeless tyre model 40...	9	33M EUR	100%
Tube tyre model 3005	4.86K	25.1M EUR	70.9%
GitiControl 278 Run-flat...	4K	15.9M EUR	64.9%
Tube tyre model 3003	4.86K	15.6M EUR	70.9%



MANUFACTURING SPC DASHBOARD

Process Control and Exception Reporting

[LEARN MORE](#)

 **plotly** | Dash

SPECIFICATION SETTINGS

CONTROL CHARTS DASHBOARD

Operator ID

1704

Time to completion

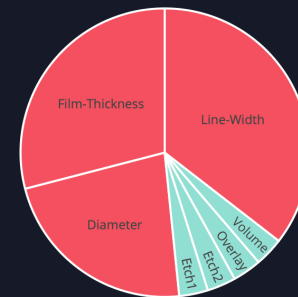
50.0

START

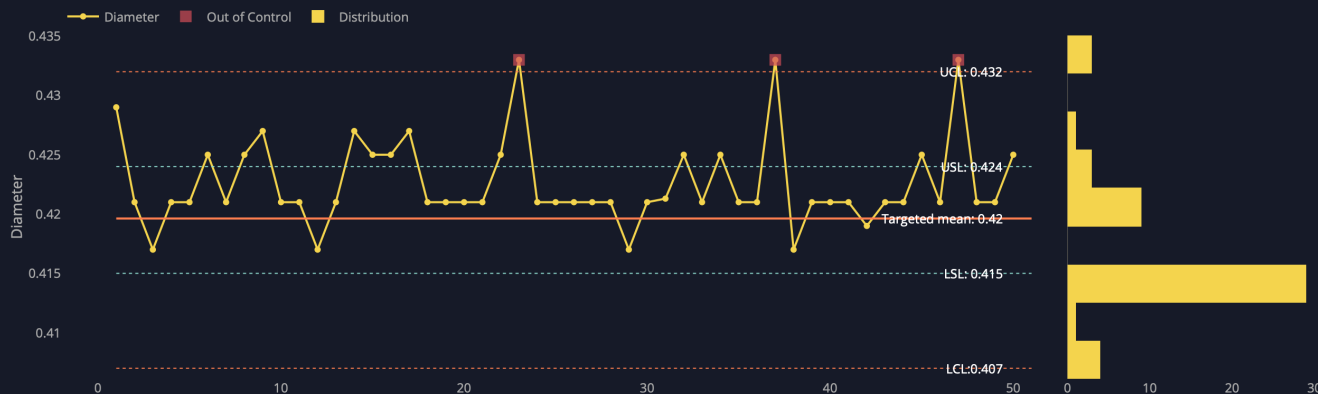
Process Control Metrics Summary

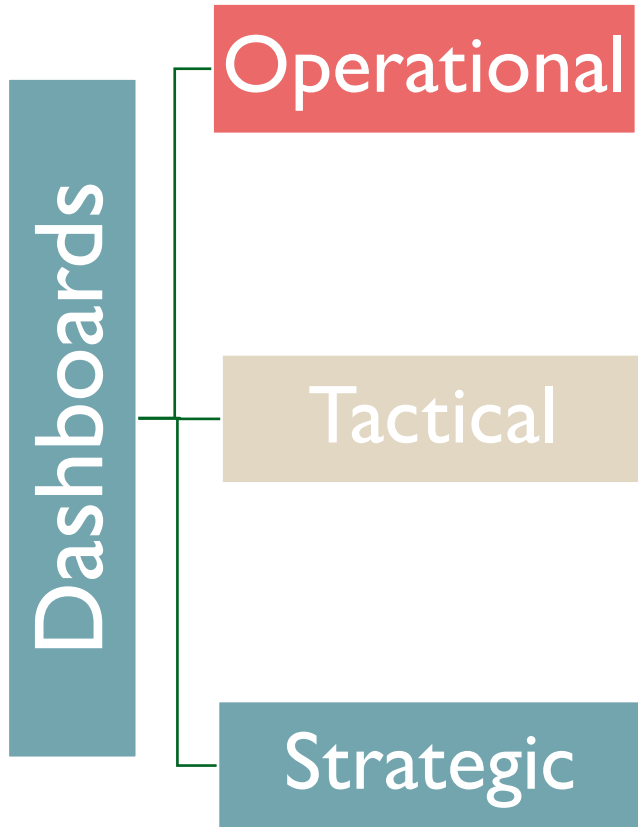
Parameter	Count	Sparkline	OOC%	%OOC	Pass/Fail
DIAMETER	50		6.00%		
ETCH1	50		0.00%		
FILM-THICKNESS	50		8.00%		
ETCH2	50		0.00%		

% OOC per Parameter



Live SPC Chart





Aimed at process workers and operational managers

- emphasis on monitoring (detect-and-respond)
 - *work-in-progress,*
 - *problematic cases (e.g., overdue/at-risk cases)*
 - *resource load*

Aimed at process owners and managers

- emphasis on analysis and management
- detecting bottlenecks
 - *cycle times*
 - *error rates*
 - *resource utilization*

Aimed at executives and managers

- emphasis on linking process performance to strategic objectives

Status Tracking

```
from event_enricher import log_track_status_count
```

```
def activity_selector(activity):
```

```
    if activity.startswith('Release'):
```

```
        return 'released'
```

```
    elif activity.startswith('Admission'):
```

```
        return 'admitted'
```

```
    elif activity == 'ER Registration':
```

```
        return 'registered'
```

```
else:
```

```
    return None
```

```
event_log = log_track_status_count(event_log,  
    status_abstraction=activity_selector)
```

defines a status for each case after each event

concept:name	init_count	registered_count	admitted_count	released_count
ER Registration	1049	1	0	0
ER Triage	1049	1	0	0
ER Sepsis Triage	1049	1	0	0
LacticAcid	1049	1	0	0
Leucocytes	1049	1	0	0
CRP	1049	1	0	0
IV Liquid	1049	1	0	0
IV Antibiotics	1049	1	0	0
Admission NC	1049	0	1	0
Leucocytes	1049	0	1	0
ER Registration	1048	1	1	0

tracks how many cases are in which state after each event

Status tracking requires monotonically increasing time

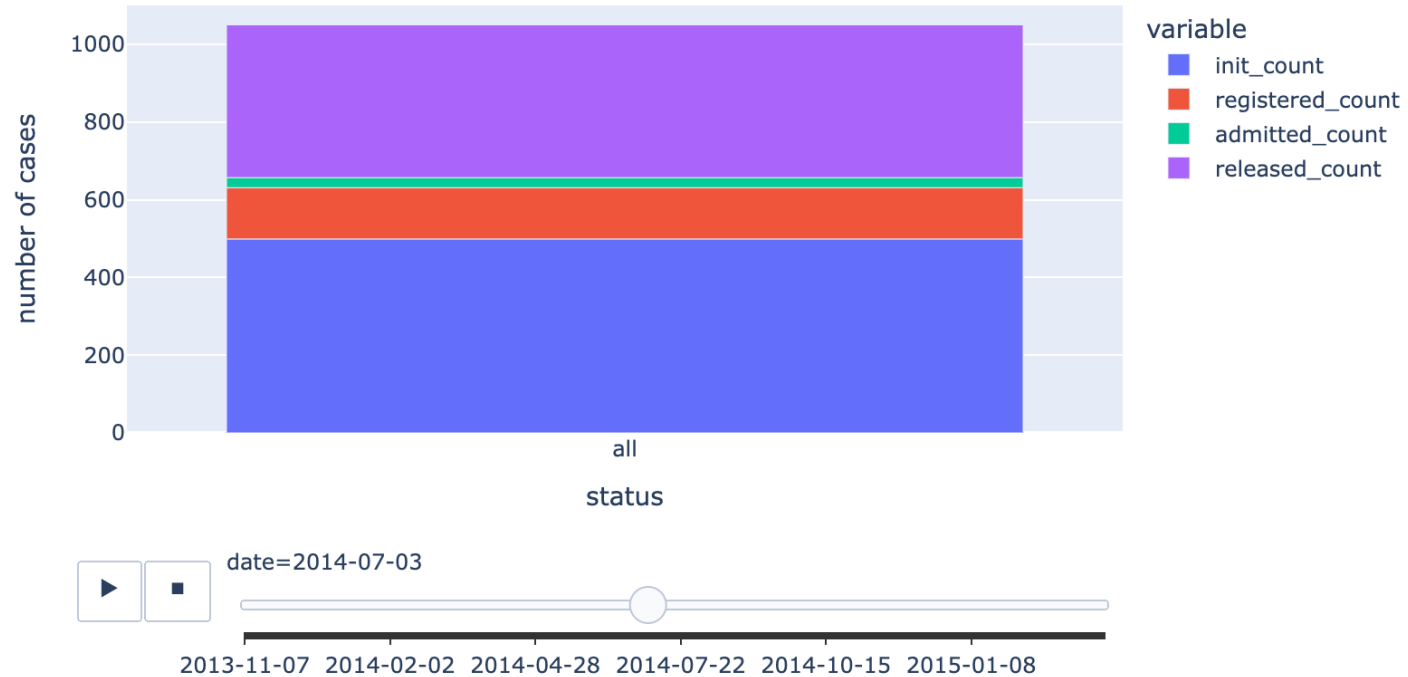
save original order

use original order as secondary ordering criterion

```
event_log['original_order'] = range(len(event_log))
event_log = event_log.sort_values(by=[COMPLETION_TIME, 'original_order'])

display(event_log[COMPLETION_TIME].is_monotonic_increasing)
```

Interactive Status Count and Animation



Discuss with your neighbor (5 mins)



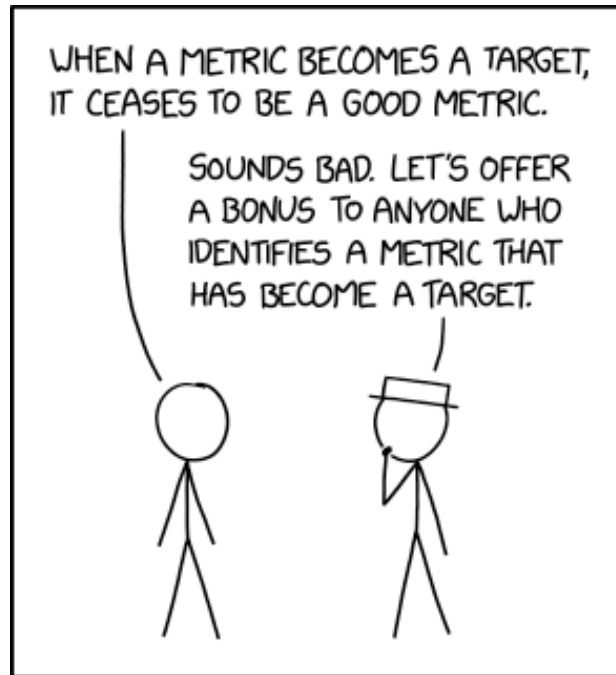
Metric-driven continuous improvement is a powerful approach – what are the traps and downsides of that approach?

Possible Guiding Questions:

- *How would you design a set of metrics to determine what is a good/improving student?*
- *As a manager, how would you design of a set of metrics to improve your team performance?*
 - *How would you communicate that to your team?*
- *Do you know an example of metric-driven improvement gone wrong?*

Performance Dashboards ...

- ... are part of [Business Activity Monitoring](#)
- are used as a driver for continuous process improvement (cf. process improvement lifecycle, Session 1)
- BPM Suites such as *Camunda* provide support
 - but there are also pandas-based libraries (e.g. [dash](#)) and many other tooling options
- *Further Reading:* W. Eckerson: [Performance Dashboards: Measuring, Monitoring, and Managing Your Business](#), Wiley 2010
- Reliance on a fixed set of metrics can be a trap
 - *Further Reading:*
 - [Goodhart's law](#)
 - Jerry Z Muller: [The Tyranny of Metrics](#)



xkcd.com

Assignment 1: Tasks from Session 3 (Outcome Analysis)

1. Define three business level process outcomes for the process under investigation.
 - a) Enrich the case log with Boolean indicators for each outcome and compute the number of cases for each outcome, for each combination of two outcomes as well as the number of cases that have no outcome defined.
 - b) Inspect and interpret a case for each of the three outcomes.
 - c) If existent, inspect and interpret a case that has more than one outcome.
 - d) If existent, inspect and interpret a case that does not belong to any of those outcomes.